POKHARA UNIVERSITY

Level: Bachelor | Semester: Fall | Year : 2023
Programme: BE | Full Marks: 100
Course: Operating System (New) | Pass Marks: 45
Time :3hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

1. a) What is operating system? Explain OS as resource manager and an extended machine. 8
   b) Producer Consumer problem can be solved with the help of various techniques. Suggest any one of the techniques to solve producer consumer with detailed explanation. 7
2. a) Given the following set of information, calculate the average waiting time and average turn-around time using FCFS, SJF, RR (Quantum = 2) and HRRN. 8

Process	Arrival Time	Service Time (Burst Time)
A	0	7
B	2	2
C	3	3
D	5	8

OR

Define threads. Explain the multithreading model in detail. 7
   b) Define PCB. Explain the Readers and Writers problem with necessary pseudocode. 8
3. a) Consider the following page reference strings: 2, 3, 4, 5, 3,2, 6,9, 3,2, 3, 4, 8,9, 4, 3,2, 3, 4,9. How many page faults would occur for each of the following page replacement algorithms assuming 3 pages a frame? In each case calculate fault ratio.
   i. FIFO page replacement
   ii. Optimal page replacement
   iii. LRU page replacement

   b) Define Thrashing. Explain the concept of paging with a suitable example. 7

OR

What is difference between Paging and Segmentation? What are different page-table structures? Explain any one.

4. a) Why is deadlock state more critical than starvation? Describe resource allocation graph with a deadlock, with a cycle but no deadlock. 7
   b) What are different ways to input output? Explain the principles of I/O software. 8
5. a) What is file? With help of necessary diagram, explain file system allocation techniques. 8
   b) Explain the concept, characteristics and role of Distributed Operating System. 7
6. a) Define pre-paging. Explain WS page replacement algorithm with example. 8
   b) Explain about Memory wall and bottleneck problem for Operating System with necessary diagrams. 7
7. Write short notes on: **(Any two)** 2×5
   a) Relocation and protection
   b) RAID
   c) Context switching

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